

THE EFFECTS OF MANUAL-HANDLING FATIGUE AND CARRIED LOAD ON TRIP RECOVERY

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INTRODUCTION

A trip occurs when a person's swing foot unexpectedly contacts an obstacle or surface. This disrupts balance and requires a rapid compensatory stepping response (CSR) to avoid a fall. [1] Trips are a leading cause of occupational falls. [2] Effective trip recovery requires a rapid and appropriate CSR to restore stability. [1] Physical fatigue has been associated with altered balance recovery following tripping [3], while anterior load carriage has also been shown to alter CSR following balance perturbation. [4] However, the combined effects of physical fatigue and anterior load carriage on the CSR during trips remain unclear. This study aims to determine whether fatigue and anterior load carriage, separately and combined, alter the CSR during trips in manual-handling workers.

METHODS

A within-participant, repeated-measures design will compare four conditions: rested/unloaded, rested/loaded, fatigued/unloaded, and fatigued/loaded. Twenty-four healthy manual-handling workers aged ≥ 18 years will be recruited. Fatigue will be induced using a standardised repeated lift-and-carry task until an RPE $\geq 15/20$ is reached, with a maximum duration of 20 minutes. In loaded conditions, participants will carry a standardised box equivalent to 15% body mass.

Participants will wear reflective markers for three-dimensional motion capture. Comfortable walking speed will be established in each condition. Each condition will conclude with an unannounced trip-like perturbation using a ground-reaction-force-triggered treadmill protocol. [5] The perturbation will be triggered during early stance when the vertical ground-reaction force of the tripped limb reaches 20-25% body weight and is increasing. [5] Perturbation magnitude will be scaled to the condition-specific walking speed. [5] The perturbed limb will be randomised and counterbalanced across conditions. Participants will wear a safety harness during the perturbation trials. A pilot phase will assess the feasibility and safety of delivering this perturbation during loaded and fatigued walking.

Primary outcomes will include recovery step length, margin of stability, step latency, and peak trunk flexion angle. Motion capture and ground reaction force data will be used to estimate joint moments during the

recovery step. A two-factor repeated-measures ANOVA will test the main effects of fatigue and load and their interaction, with the fatigue $\times$ load interaction as the primary effect of interest.

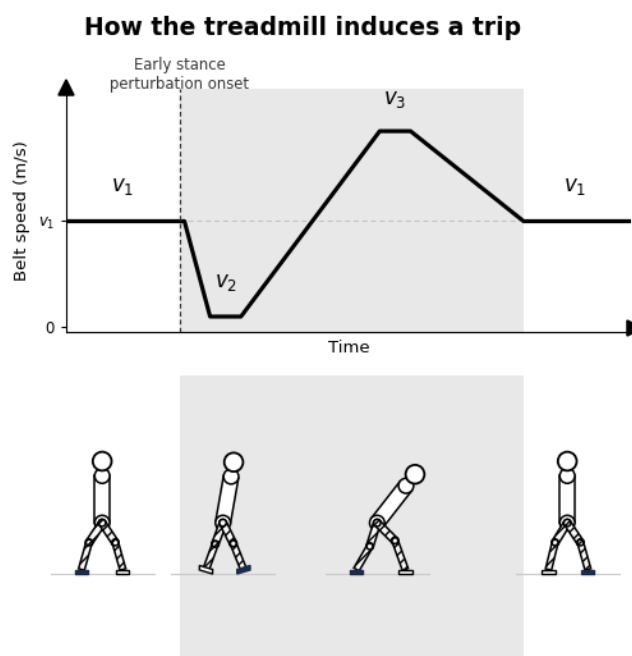


Figure 1. Schematic of the treadmill-based trip-like perturbation used during walking.

EXPECTED OUTCOMES AND SIGNIFICANCE

The combined fatigue and load condition is expected to produce the greatest impairment in the CSR. This may indicate reduced capacity to recover from trips during combined exposure. If the combined effect is greater than the effects of either condition alone, findings could support workplace strategies that address both exposures together. If the effect is primarily attributable to fatigue or load carriage alone, findings could inform fatigue-management or load-management strategies, respectively.

REFERENCES

- [1] Sessoms PH et al. *J Biomech* **47**: 277–280, 2014.
- [2] Qu X et al. *Hum Factors* **62**: 919–927, 2020.
- [3] Allin LJ, Madigan ML. *IIEE Trans Occup Ergon Hum Factors* **8**: 155–165, 2020.
- [4] Marchant HJ, Zettel JL. *J Appl Biomech* **42**: 336–342, 2026.
- [5] Shih H-T et al. *PLoS One* **18**: e0284384, 2023.